POWERACT CONSULTING

NCP Solver

DynamicMS Suite

SOFTWARE REQUIREMENTS SPECIFICATION

Functional & Non-Functional Requirements, and Interfaces

Version 1.5

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Confidential

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) captures the complete set of functional and non-functional requirements, and the external interfaces, of NCP Solver — DynamicMS Suite. It is derived from, and traceable to, the NCP Solver project scope, definition & design (PDD), information model, and knowledge-base source specifications, and from the delivered implementation itself. It is intended for product owners, developers, quality assurance, and auditors.

1.2 Scope

NCP Solver is a full-stack, multi-tenant, multi-language web application that manages non-conformities and problems (recorded as "NCP Sheets") through a structured seven-stage resolution process (S1-S7), augmented by rule-based AI agents, and surrounded by governance, risk, and compliance (GRC) modules: Business Rules, Controls (COSO framework), Risks & Opportunities, a RACSI (Responsible/Accountable/Consulted/Support/Informed) accountability matrix, and a BPMN process-diagram module, plus an independently versioned AI Use Cases Library. An Organizational Breakdown Structure (OBS) of sites/departments/services/teams underlies and links every one of these modules. A chat-style AI Assistant answers RBAC-scoped data questions and application-help questions, and a Governance-group screen lets an administrator configure a Large Language Model (LLM) provider for future generative features. An in-app, searchable, multi-language Help guide documents every module. The application supports both independent-company and group-of-companies tenancy models, role-based access control across nine standard roles, and English/French/Arabic localization including full right-to-left layout.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
NCP Sheet	The central record of a non-conformity or problem, spanning stages S1-S7 (referred to as "NCP Fiche" in the original source specification; stages were renumbered from E1-E7 to S1-S7 in the delivered implementation).
S1-S7	The seven stages of the NCP resolution process: Detection & Alert, Problem Understanding, Immediate/Containment Actions, Root Cause Analysis, Corrective Action Plan, Execution & Evaluation, Capitalization (REX).
AR	Action Responsible (Action Owner) - the user responsible for executing an action. Renamed from RR (Responsable Réalisation) in the delivered implementation.
AE	Action Evaluator - the user responsible for verifying an action's effectiveness; must differ from the AR. Renamed from RE (Responsable Évaluation) in the delivered implementation.
REX	Retour d'EXperience - the structured lessons-learned output of stage S7.

Term	Definition
RCA	Root Cause Analysis - the discipline of finding the true underlying cause of a problem (5-Why, Ishikawa).
5W2H / QQOQCCP	Structured problem-description framework: What/Where/When/Who/Why/How/How-Much (French: Qui/Quoi/Ou/Quand/Comment/Combien/Pourquoi).
6M	Ishikawa cause categories: Man, Machine, Method, Material, Measurement, Milieu.
RAG	Retrieval-Augmented Generation - TF-IDF/cosine-similarity semantic search grounding AI suggestions and the AI Assistant's application-help answers in the tenant's own historical data or the built-in FAQ corpus, without a live external LLM call.
RBAC	Role-Based Access Control - access rights determined by assigned role(s).
COSO	Committee of Sponsoring Organizations of the Treadway Commission - the internal control framework used to classify Controls.
OBS	Organizational Breakdown Structure - the site/department/service/team tree; the shared source of roles and named people that other modules, including RACSI, point back to.
RACSI	Responsible, Accountable, Consulted, Support, Informed - a five-role extension of the classic RACI accountability model; exactly one Accountable per activity, the other four roles allow multiple assignees.
BPMN	Business Process Model and Notation - the OMG standard notation used by the BPMN Process module to model workflows, including the NCP Solver process itself.
LLM	Large Language Model - the class of AI model an AI Use Case may optionally call via a user's own Real LLM Provider Connection (section 3.11), stored only in that user's browser and forwarded once per generation request; every AI Use Case still computes its deterministic result and falls back to it automatically if the live call fails.
AI Assistant	The chat-style Q&A feature (section 3.23) offering RBAC-scoped data queries and application-help queries, distinct from both the per-stage AI Agents and the AI Use Case Library.
Assistive / Augmented	The two AI Use Case tiers (section 3.11): Assistive means the AI only observes/analyzes/suggests while a human performs the task and decision; Augmented means the AI performs a substantial part of the task but a human must review, edit, and approve before it is finalized. A third tier, Autonomous, is explicitly out of scope.
Multi-Tenant	Architecture in which multiple Organizations share one deployment with fully isolated data.
Composable Module	A module (Action Register, Alert Engine, RBAC, Audit Trail) built for reuse by other applications via a shared API contract.

1.4 References

- NCP Solver - Fast-Track Project Scope (source specification).
- NCP Solver - AI Project Definition & Design (PDD) v1.0.
- NCP Solver - Information Model (core classes / entity specification).
- NCP Solver - Knowledge Base for RAG v1.0 (KB-000 through KB-021).
- NCP-Solver: Global CIO Forum Use Case presentation, Fast-Track edition.

- POWERACT Consulting - Brand & Graphical Chart Guide v1.0.

1.5 Document Overview

Section 2 describes the product at a high level. Section 3 enumerates functional requirements by module, each individually identified (FR-xxx) for traceability. Section 4 defines external interfaces (user, software, hardware, communications, and interfaces with other applications). Section 5 defines non-functional requirements (NFR-xxx). Section 6 summarizes the data model. Appendices provide the permission matrix, KPI formulas, alert catalog, and a full glossary.

2. Overall Description

2.1 Product Perspective

NCP Solver is a new, self-contained system composed of a React single-page-application frontend and a Node.js/Express REST API backend, persisting to a relational database. It is designed around six composable modules (Action Register, Capitalization Library/RAG, Alert & Notification Engine, User Management & RBAC, KPI Dashboard & Reporting, Audit Trail Service) intended for reuse by sibling enterprise applications (Risk Management, Audit Management, Project Management) via the same API contract.

2.2 Product Functions (Summary)

- Register, understand, contain, analyze, correct, execute/evaluate, and capitalize non-conformities via the S1-S7 NCP Sheet workflow.
- Manage an Action Register shared by immediate and corrective actions, with AR/AE-segregated (Action Responsible / Action Evaluator) effectiveness evaluation and evidence.
- Provide one rule-based AI agent per NCP process step (S1-S7) plus an always-on Monitoring & Alert Agent, that suggest — never autonomously decide — classifications, problem structuring, containment actions, root causes, corrective actions, evaluation context, and REX narratives.
- Maintain a semantically searchable Capitalization Library (RAG) of resolved problems, tenant-isolated.
- Maintain an independently versioned AI Use Cases Library with full labeled-section content history, revert, and an Active/Inactive toggle.
- Maintain Business Rules, COSO-classified Controls, and a Risk/Opportunity register with a likelihood x impact heat map.
- Maintain a RACSI accountability matrix — one Accountable, any number of Responsible/Consulted/Support/Informed, as roles or named people — per NCP process step and per linked governance item.
- Model process workflows, including the NCP Solver process itself, as real BPMN 2.0 diagrams, viewable or editable in-browser based on RBAC.
- Provide an AI Assistant answering RBAC-scoped questions about the organization's own data and "Can it / How do I" questions about the application itself, open to every role.
- Provide a Governance-group screen to configure a Large Language Model (LLM) provider, from a predefined list of the ten most common providers, as the designated integration point for future generative features.
- Provide full role-based access control across 9 standard roles and a live permission matrix editor.
- Provide Group/Organization/Project hierarchy and an Organizational Breakdown Structure (OBS) that links to Sheets, Business Rules, Controls, Risks/Opportunities, RACSI activities, BPMN

diagrams, and user role assignments, with live linked-item counts per node.

- Compute 10 KPIs and 4 standard reports, and raise 10 categories of alerts.
- Operate multi-tenant with complete data isolation, and multi-language with full Arabic RTL support.
- Provide a searchable, multi-language in-app Help guide covering every module, open to all authenticated users.

2.3 User Classes and Characteristics

Role	Who They Are	Key Permissions
Administrator	IT/system administrator	Full CRUD on all records; user & RBAC management; tenant configuration; license, governance, and full AI Use Case Library governance (catalog CRUD, Organization activation, Project overrides, usage log).
Quality Manager	Owns the quality management system	Validates and closes Sheets; configures KPI thresholds; owns Controls and Business Rules; full CRUD on the RACSI matrix; compliance reporting; full AI Use Case Library governance (catalog CRUD, Organization activation, Project overrides, usage log).
CI Pilot	Coordinates the NCP process for their scope	Creates/edits/validates Sheets; approves action plans; manages REX; owns Business Rules and Risks/Opportunities in scope; creates/edits RACSI activities and assignments; views BPMN diagrams; catalogs and edits AI Use Cases, sets Project-level AI overrides, and views the AI Usage Log.
NCP Team Member	Cross-functional problem-solving team	Views/edits assigned Sheets; completes S2/S4/S5 stages; adds root causes and actions.
Action Owner (AR)	Executes an assigned action	Updates own action status; enters completion date; uploads execution evidence.
Evaluator (AE)	Verifies action effectiveness	Views pending evaluations; records effectiveness verdict; uploads evaluation evidence. Must differ from the action's Action Owner (AR).
Department Head	Manages an affected department	Views all Sheets in own department; department dashboard/reports; receives department-level alerts, risk visibility, RACSI matrix visibility, and BPMN view access.
Reporter	Any employee authorized to report a problem	Creates new Sheets (S1 only); views own submitted Sheets; no edit rights after submission.
Read-Only / Auditor	Internal or external audit/compliance	Read-only access to Sheets, reports, Business Rules, Controls, Risks/Opportunities, the RACSI matrix, BPMN diagrams, the AI Use Case Library and its AI Usage Log, and the full audit trail.

2.4 Operating Environment

Server: any standard Node.js 22.5+ runtime (Node 22 LTS recommended), no external database server, third-party API key, or native/compiled build toolchain required (the relational layer uses Node's built-in node:sqlite module; the relational and RAG layers are each swappable for a distributed equivalent). Client: any evergreen desktop or tablet web browser. Deployment: SaaS (shared multi-tenant infrastructure) or OnPrem (customer-hosted), selectable per Organization.

2.5 Design and Implementation Constraints

- All AI agent behavior must be explainable and logged (agent name, input, output, confidence score); no black-box autonomous decisions.
- Every data access must be scoped by tenant (organization_id); no query may return another tenant's data.
- The UI must conform to the POWERACT Consulting visual identity across every module and every supported language.
- The Action Owner (AR) and Evaluator (AE) on any single action evaluation must always be different users.
- The AI Assistant's data-query mode must check the matched intent's required permission before running any query, and must refuse transparently — never partially or silently — when that permission is absent.
- A stored LLM provider API key must never be returned by any API response once saved, mirroring the existing password-hash-never-returned pattern.

2.6 Assumptions and Dependencies

- Users have modern web browsers with JavaScript enabled.
- Organizations are responsible for their own user provisioning within the tenant they administer.
- The rule-based AI agent implementations are a functional stand-in for a future LLM/vector-database-backed implementation, behind a stable interface.

3. System Features (Functional Requirements)

Each requirement is uniquely identified and stated in "the system shall" form for verifiability. Requirements are grouped by module in implementation order.

3.1 Dashboard

A role-aware landing page summarizing the authenticated user's open work, recent NCP Sheets, and unread alerts.

ID	Requirement
FR-DASH-01	The system shall display, on login, counts of NCP Sheets by status (Open, In Progress, Closed, Cancelled) scoped to the user's organization.
FR-DASH-02	The system shall display the count of the authenticated user's own open assigned actions and the count of organization-wide overdue actions.
FR-DASH-03	The system shall list the most recently created NCP Sheets, each showing sheet number, title, stage, criticality, and status.
FR-DASH-04	The system shall list the authenticated user's unread alerts with type, message, and timestamp.
FR-DASH-05	The system shall render a bar chart of non-conformity counts grouped by department (OBS node).
FR-DASH-06	The system shall provide a one-click action to create a new NCP Sheet, visible only to users holding the fiche.create permission.

3.2 Hierarchy Management (Group / Organization / Project)

Supports an optional Group (holding company) above the Organization (tenant) level, and optional Projects below it, per the Group-of-companies and independent-company deployment scenarios.

ID	Requirement
FR-HIER-01	The system shall allow an Organization to optionally belong to a Group; an Organization not assigned to a Group shall operate as fully independent.
FR-HIER-02	When an Organization belongs to a Group, the system shall display all sibling Organizations in that Group to authorized users.
FR-HIER-03	The system shall allow an authorized user to create an additional sibling Organization within the same Group.
FR-HIER-04	The system shall allow full CRUD on Projects (optional sub-entities of an Organization), including name, description, status, and start/end dates.
FR-HIER-05	The system shall allow an NCP Sheet to be optionally associated with a Project.
FR-HIER-06	The system shall allow an authorized user to edit core Organization attributes (name, sector, country).

3.3 Organizational Breakdown Structure (OBS)

A hierarchical site/department/service/team tree used to classify where a non-conformity occurred, and the shared "who/where" backbone that Business Rules, Controls, Risks & Opportunities, RACSI activities, BPMN diagrams, and user role assignments all point back to.

ID	Requirement
FR-OBS-01	The system shall support full CRUD on OBS nodes, each typed as Site, Department, Service, or Team.
FR-OBS-02	The system shall allow an OBS node to have a parent OBS node, forming an arbitrary-depth tree per Organization.
FR-OBS-03	The system shall render the OBS as an indented tree reflecting parent/child relationships.
FR-OBS-04	The system shall allow an NCP Sheet to be associated with exactly one OBS node (its department/origin).
FR-OBS-05	The system shall prevent deleting an OBS node from breaking referential integrity by cascading or reassigning dependent records per the data model.
FR-OBS-06	The system shall allow a Business Rule, a Control, a Risk/Opportunity, a RACSI Activity, and a BPMN Diagram to each be optionally tagged with the OBS node that owns them.
FR-OBS-07	The system shall allow a user's role assignment to be scoped to a specific OBS node, giving that node a "home" set of people in addition to organizational units.
FR-OBS-08	The system shall display, for every OBS node, a live count of every linked item type (people, NCP Sheets, Business Rules, Controls, Risks & Opportunities, RACSI Activities) so accountability and activity concentration are visible at a glance.

3.4 Identity & Role-Based Access Control (RBAC)

User directory, the 9 standard roles from the NCP Solver Knowledge Base (KB-011), and a full permission matrix editor.

ID	Requirement
FR-ID-01	The system shall support full CRUD on users, including first/last name, email, username, language preference, and active status.
FR-ID-02	The system shall hash all user passwords (bcrypt) and never expose password hashes via any API response.
FR-ID-03	The system shall support assigning one or more of the 9 standard roles to a user: Administrator, Quality Manager, CI Pilot, NCP Team Member, Action Owner (AR), Evaluator (AE), Department Head, Reporter, Read-Only/Auditor.
FR-ID-04	The system shall support full CRUD on custom roles per Organization, in addition to the 9 system role templates.
FR-ID-05	The system shall provide a Permission Matrix screen displaying every permission code against every role, with checkboxes editable by users holding role.managePermissions.
FR-ID-06	The system shall enforce every API endpoint against the authenticated user's resolved permission set (union across all assigned roles); a request lacking the required permission shall be rejected with HTTP 403.

ID	Requirement
FR-ID-07	The system shall enforce that an Evaluator (AE) recording an action's effectiveness verdict is never the same user as that action's Action Owner (AR).
FR-ID-08	The system shall provide a minimal user directory (id, name, email) accessible to any authenticated user for assigning owners/evaluators/team members, independent of the full user.view permission.
FR-ID-09	The system shall issue a signed JSON Web Token (JWT) on successful login, valid for 12 hours, required as a Bearer token on every subsequent API call.

3.5 Governance Settings

Tenant-level configuration of process defaults, KPI thresholds, and alert behavior.

ID	Requirement
FR-GOV-01	The system shall allow an authorized user to configure the default Root Cause Analysis method (5-Why or Ishikawa) for the Organization.
FR-GOV-02	The system shall allow an authorized user to toggle whether a completed REX (S7) entry is required before an NCP Sheet can be closed.
FR-GOV-03	The system shall allow an authorized user to configure the numeric threshold for each of the 10 KPIs.
FR-GOV-04	The system shall allow an authorized user to enable or disable each of the 10 alert types (A-J) independently.
FR-GOV-05	The system shall persist governance settings per Organization and apply tenant-specific defaults automatically on first access.

3.6 License & Plan Management

SaaS/OnPrem deployment tracking and seat/plan administration.

ID	Requirement
FR-LIC-01	The system shall record, per Organization, a plan tier (Starter, Professional, Enterprise) and a deployment model (SaaS or OnPrem).
FR-LIC-02	The system shall track total seats, seats used (computed from active user count), billing cycle, renewal date, and license status.
FR-LIC-03	The system shall allow an authorized user to edit the plan tier, deployment model, seat allocation, and billing cycle.
FR-LIC-04	The system shall recompute seats-used automatically whenever the active user count changes.

3.7 NCP Sheet Lifecycle (S1 through S7)

The central workflow: a structured, staged record of a non-conformity or problem from detection through capitalization, matching KB-003 through KB-009. Stages were renamed from the original specification's E1-E7 numbering to S1-S7 in the delivered implementation; Action Owner/Evaluator were correspondingly renamed from RR/RE to AR/AE (Action Responsible / Action Evaluator).

ID	Requirement
FR-S1-01	The system shall allow any user holding fiche.create to register a new NCP Sheet capturing title, description, detection date, criticality (High/Medium/Low), priority (P1-P3), frequency (first-time/recurring), and applicable standard.
FR-S1-02	The system shall auto-generate a unique, tenant-scoped Sheet Number in the format NC-{year}-{4-digit sequence} on creation.
FR-S1-03	The system shall set a newly created Sheet's stage to S1 and status to Open.
FR-S1-04	The system shall invoke the Classification Agent on Sheet creation, logging a suggested criticality, suggested priority, and a list of similar past Sheets with a confidence score.
FR-S1-05	The system shall allow the Classification Agent to be re-run on demand from the Sheet's S1 tab.
FR-S1-06	The system shall raise Alert A immediately when a new Sheet is created with criticality High or priority 1.
FR-S2-01	The system shall provide a structured 5W2H (What/Where/When/Who/Why/How/How-Much) form for the Problem Understanding stage (S2), plus a free-text Objectives field.
FR-S2-02	The system shall persist exactly one Problem Understanding record per Sheet, editable by users holding fiche.edit.
FR-S2-03	The system shall invoke the Problem Structuring Agent on request during S2, suggesting a structured 5W2H draft from the Sheet's title and description to accelerate the team's analysis.
FR-S3-01	The system shall allow one or more Immediate Actions to be created against a Sheet, each with description, tasks, required means, an Action Owner (AR), and planned completion date.
FR-S3-02	The system shall allow the Immediate Action's Action Owner (AR) to mark it Done and record the actual completion date.
FR-S3-03	The system shall allow an Evaluator (AE), who must differ from the action's Action Owner (AR), to record an effectiveness verdict (Effective / Not Effective) with comments and efficiency criteria for each Immediate Action.
FR-S3-04	The system shall allow one or more evidence attachments (execution proof / evaluation proof) to be recorded against any action.
FR-S3-05	The system shall invoke the Containment Advisor Agent on request, returning suggested immediate actions drawn from similar closed Sheets.
FR-S3-06	The system shall raise Alert B when an Immediate Action's planned completion date has passed and its status is not Done, and Alert G one day before that date.
FR-S4-01	The system shall allow one or more Root Causes to be recorded against a Sheet, each with a description, a 6M cause category (Man/Machine/Method/Material/Measurement/Milieu), and an RCA method (5-Why or Ishikawa).
FR-S4-02	The system shall allow a Root Cause to be optionally linked to a Standard.
FR-S4-03	The system shall invoke the Root Cause Mining Agent on request, returning suggested root causes and 6M categories drawn from similar closed Sheets.
FR-S4-04	The system shall raise Alert D when 48 hours have elapsed since detection and no Root Cause has yet been recorded for the Sheet.

ID	Requirement
FR-S5-01	The system shall allow one or more Corrective Actions to be defined per Root Cause, each with description, tasks, required means, an Action Owner (AR), and planned completion date.
FR-S5-02	The S5 (Corrective Action Plan) view shall present action definition only; execution status and effectiveness controls shall not be exposed on this tab.
FR-S5-03	The system shall invoke the Action Recommendation Agent on request, returning proven corrective actions for the given root cause(s) drawn from the Capitalization Library.
FR-S6-01	The S6 (Execution & Evaluation) view shall present the same Corrective Actions defined in S5 with execution controls: mark Done, attach evidence, and record an AR/AE-separated effectiveness verdict.
FR-S6-02	The system shall raise Alert C when a Corrective Action is marked Done or its evaluation deadline is 2 days away, and Alert H two days before its planned completion date.
FR-S6-03	The system shall raise Alert I two days before a scheduled evaluation's planned review date.
FR-S6-04	The system shall invoke the Evaluation Assistant Agent on request during S6, surfacing, for each pending Corrective Action, the specific root cause to re-verify and the organization's historical corrective-action effectiveness rate as evaluation context.
FR-S7-01	The system shall provide a REX (Retour d'EXpérience) form capturing Lessons Learned, Root Cause Summary, Solution Summary, a Standardization decision (yes/no + details), a Generalization decision (yes/no + plan), and searchable tags.
FR-S7-02	The system shall invoke the REX Generation Agent on request, auto-drafting the Lessons Learned narrative and standardization/generalization recommendations from the Sheet's full S1-S6 content.
FR-S7-03	The system shall raise Alert E when all Corrective Actions on a Sheet have an Effective verdict, prompting the team to complete S7.
FR-S7-04	The system shall block closing a Sheet (transition to status Closed) unless a REX entry exists, when the Organization's Governance Settings require it (default: required).
FR-S7-05	Upon closure, the system shall set the Sheet's status to Closed, stage to S7, and closure date to the current date, and shall make the Sheet's content available to the Capitalization Library RAG index.
FR-GEN-01	The system shall provide an explicit, sequential stage-advance action (S1->S2->...->S7) gated by fiche.validate, and shall track the Sheet's current stage independently of its status.
FR-GEN-02	The system shall raise Alert J when a Priority-1 Sheet has had no update for 3 or more consecutive days.
FR-GEN-03	The system shall allow assignment of one or more NCP Team Members to a Sheet, distinct from its original detector.

3.8 Action Register

A composable, reusable module underlying both Immediate (S3) and Corrective (S5/S6) actions.

ID	Requirement
FR-ACT-01	The system shall auto-generate a unique, tenant-scoped Action Number in the format A-{year}-{4-digit sequence} on creation.
FR-ACT-02	The system shall track an Action's status through To Do -> In Progress -> Done -> Cancelled.

ID	Requirement
FR-ACT-03	The system shall provide a personal "My Actions" view listing actions owned by, or pending evaluation by, the authenticated user.
FR-ACT-04	The system shall restrict an action's self-service status update to its assigned Action Owner (AR) (or a user holding action.edit).

3.9 Capitalization Library & Retrieval-Augmented Generation (RAG)

The organizational memory of resolved problems, semantically searchable.

ID	Requirement
FR-CAP-01	The system shall automatically index every closed Sheet's title, description, root causes, corrective actions, and REX content into a per-tenant search index upon closure.
FR-CAP-02	The system shall provide a free-text semantic search over the Capitalization Library, ranked by similarity score, restricted to the requesting user's own tenant.
FR-CAP-03	The system shall display, for each closed Sheet in the library, its lessons learned, standardization flag, and generalization flag.

3.10 AI Agents (Process Assistance)

Rule-based, explainable assistants providing suggestions, never autonomous decisions: one for every NCP process step (KB-010) plus an always-on Monitoring & Alert Agent. Every one of these agents is itself a governed entry in the AI Use Case Library (section 3.11), which controls its activation, discloses its RAG grounding, and logs every outcome. A separate pair of AI Assistant agents (RBAC-scoped data query and application help query) is specified in section 3.23.

ID	Requirement
FR-AI-01	The system shall log every AI agent invocation (agent name, prompt/context, response, confidence score, timestamp, related Sheet) to an immutable AI Agent Log.
FR-AI-02	The system shall never allow an AI agent to autonomously change a Sheet's data; every suggestion shall require explicit human action to apply.
FR-AI-03	The system shall scope every AI agent's retrieval to the requesting user's own tenant (no cross-tenant data in suggestions).
FR-AI-04	The system shall implement the Monitoring & Alert Agent as an always-on process computing Alerts A-J from current data, invocable on demand or on a schedule.
FR-AI-05	The system shall provide one dedicated agent per NCP process step — Classification (S1), Problem Structuring (S2), Containment Advisor (S3), Root Cause Mining (S4), Action Recommendation (S5), Evaluation Assistant (S6), and REX Generation (S7) — so every stage of the workflow, not only a subset, carries AI assistance.

3.11 AI Use Case Library & Governance

The governed catalog and control plane for every AI capability in NCP Solver, independent of the Capitalization Library. Every use case sits in exactly one of two tiers — Assistive (the AI observes, analyzes, or suggests; a human *performs the task and makes the decision) or Augmented (the AI performs a substantial part of the task — drafting, classifying, or summarizing at scale — but a human must review, edit, and approve the result before it is finalized or distributed). A third tier, Autonomous AI, in which the system would take an irreversible action without a human decision in the loop, is deliberately out of scope for this release and would require a separate, explicitly-gated tier in a future one. Each Organization is seeded with fourteen versioned AI Use Cases: the seven built-in process agents (S1-S7, section 3.10), the two AI Assistant modes (section 3.23), and five forward-looking, catalog-only entries (BPMN diagram documentation, risk narrative drafting, control test summarization, business rule drafting, and RACSI assignment recommendation) recorded at an early maturity stage with no live invocation, so the catalog honestly reflects what is shipped versus planned. The Library also hosts the optional Real LLM Provider Connection that every live AI Use Case can draw on (FR-AIUC-17 to FR-AIUC-19).*

ID	Requirement
FR-AIUC-01	The system shall support full CRUD on AI Use Cases, capturing an id, name, tier (Assistive/Augmented), owning module, description, trigger condition, expected output, business function, AI technique, maturity stage (1-5), status (Idea/Pilot/Production/Retired), expected impact, estimated ROI, and tags.
FR-AIUC-02	The system shall capture, per AI Use Case, a trigger description (what causes it to run), an output description (what it produces), and a human checkpoint description (the point at which a person must review, edit, or approve its output before it has any effect) — distinct fields from the versioned prompt-template content of FR-AIUC-03.
FR-AIUC-03	The system shall version an AI Use Case's editable prompt-template content independently of its metadata; content shall consist of five labeled sections: Inputs, Prompt, Expected Output, Constraints/Guardrails, and Model/Technique Notes.
FR-AIUC-04	The system shall create Version 1 of an AI Use Case automatically upon creation and mark it as the Default and Current version.
FR-AIUC-05	The system shall allow an authorized user to save the current draft as a new version, incrementing the version number and marking it Current; the prior version shall remain in history unmodified.
FR-AIUC-06	The system shall display the full Version History for an AI Use Case, showing version number, author, timestamp, change note, and Default/Current status badges, and shall allow viewing the full labeled content of any historical version.
FR-AIUC-07	The system shall allow an authorized user to revert to any historical version; reverting shall duplicate that version's content into a brand-new version marked Current, never rewrite or delete history.
FR-AIUC-08	The system shall gate full CRUD on AI Use Cases and their version history — create, metadata edit, delete, and version revert — behind a single canManageAiUseCases capability (aiUseCase.create / aiUseCase.edit / aiUseCase.delete), distinct from the activation capabilities of FR-AIUC-11 and FR-AIUC-12.
FR-AIUC-09	The system shall seed every Organization, on creation, with the same fourteen AI Use Cases described in the section intro, so the catalog is identical in shape across every tenant regardless of sector.
FR-AIUC-10	The system shall allow an Organization Administrator holding canActivateAiForOrg (aiUseCase.activate) to activate or deactivate any AI Use Case for the whole Organization, independently of editing its metadata or content.
FR-AIUC-11	The system shall allow a user holding canRequestProjectAiOverride (aiUseCase.projectOverride) to set a per-Project override on any AI Use Case as one of three states: Inherit, On, or Off.
FR-AIUC-12	The system shall compute an AI Use Case's effective activation state, for a given Project context, as that Project's override when set to On or Off, and as the Organization's activation state (FR-AIUC-10) when

ID	Requirement
	the Project override is Inherit or unset.
FR-AIUC-13	The system shall block, server-side, any new AI suggestion from a deactivated AI Use Case (by its effective state) immediately upon deactivation, while never retroactively hiding, unlabeling, or removing content a human already reviewed and approved before deactivation.
FR-AIUC-14	The system shall visually label every AI-generated output, wherever it is shown, as AI-generated and requiring human review, alongside its tier (Assistive/Augmented).
FR-AIUC-15	The system shall write an append-only AI Usage Log entry for every AI suggestion's outcome — Accepted, Edited, or Rejected — capturing the AI Use Case, Organization, Project (if any), the record it applied to, an output summary, the acting user, and a timestamp; the system shall expose no edit or delete route or UI control for this log.
FR-AIUC-16	The system shall, on deleting an AI Use Case, purge every Organization activation entry and every Project-override entry that reference it, so no orphaned activation-map key can outlive its use case.
FR-AIUC-17	The system shall allow a user to configure a Real LLM Provider Connection (provider, model, endpoint URL, and API key) and shall persist it only in the user's own browser local storage, in a namespace separate from every other cached or exportable application state.
FR-AIUC-18	The system shall never include the Real LLM Provider Connection, in whole or in part, in any data export, and shall never synchronize, cache, or persist it on the backend beyond the single generation request it is forwarded with; the connection shall survive a Reset Demo Data operation, since that operation resets only server-side tenant data.
FR-AIUC-19	The system shall, on any failure of a live LLM call — network error, authentication failure, timeout, or an unsupported provider — fall back automatically to its deterministic built-in generator for that same request, so a misconfigured or unreachable connection never blocks a suggestion from being produced.
FR-AIUC-20	The system shall ensure an AI Use Case's activation, tier, or generation never grants a user visibility into any data beyond what their RBAC role and permissions already allow.
FR-AIUC-21	The system shall, before generating any suggestion, retrieve the top-matching methodology reference definitions (Standards) for the record at hand via its RAG engine, instruct the underlying model (real or deterministic) to stay strictly within NCP Solver's own stage and category vocabulary (S1-S7, AR/AE, the 6M causes) and to state no number, score, or name that is not present in the retrieved references or the record's own data, and disclose the retrieved definitions to the user alongside the output.

3.12 Business Rules

A governed catalog of the process rules that constrain NCP Solver's own behavior, with full CRUD via RBAC.

ID	Requirement
FR-BR-01	The system shall support full CRUD on Business Rules, each with a code, title, rule type (Validation/Workflow/Approval/Naming/Threshold/Escalation), the module it applies to, a Condition (IF) statement, an Action (THEN) statement, severity (Blocking/Warning/Info), and an owner.
FR-BR-02	The system shall allow a Business Rule to be marked active or inactive without deleting it.
FR-BR-03	The system shall restrict create/edit/delete of Business Rules to users holding the corresponding businessRule permission.

ID	Requirement
FR-BR-04	The system shall allow a Business Rule to be tagged with the owning OBS node and, via the RACSI matrix, with a named Accountable owner.

3.13 Controls (COSO Framework)

Internal controls mapped to the five COSO Internal Control - Integrated Framework components, with full CRUD via RBAC.

ID	Requirement
FR-CTL-01	The system shall support full CRUD on Controls, each classified under one of the five COSO components: Control Environment, Risk Assessment, Control Activities, Information & Communication, or Monitoring Activities.
FR-CTL-02	The system shall record, per Control, a control type (Preventive/Detective/Corrective), a testing frequency (Continuous through Annual), an owner, an effectiveness rating (Effective/Partially Effective/Ineffective/Not Tested), last-tested and next-test-due dates, and evidence notes.
FR-CTL-03	The system shall allow filtering the Controls list by COSO component.
FR-CTL-04	The system shall allow a Control to be linked to one or more Risks/Opportunities as a mitigating control.
FR-CTL-05	The system shall allow a Control to be tagged with the owning OBS node and, via the RACSI matrix, with a named accountability structure for its periodic testing.

3.14 Risks & Opportunities

A likelihood x impact risk register covering both risks and opportunities related to operating NCP Solver, with full CRUD via RBAC.

ID	Requirement
FR-RISK-01	The system shall support full CRUD on Risk/Opportunity items, each with a code, title, description, type (Risk/Opportunity), category (Strategic/Operational/Compliance/Financial/Reputational/Technology), likelihood (1-5), impact (1-5), response strategy, mitigation/action plan, owner, and status (Identified/Assessing/Mitigating/Monitoring/Closed).
FR-RISK-02	The system shall automatically compute an Inherent Score (likelihood x impact) and, where residual likelihood/impact are set, a Residual Score.
FR-RISK-03	The system shall allow a Risk/Opportunity to be linked to zero or more mitigating Controls.
FR-RISK-04	The system shall render a 5x5 likelihood x impact heat-map matrix, color-banded from lowest risk (dark green) to highest risk (red), plotting every Risk item in its corresponding cell.
FR-RISK-05	The system shall allow filtering the register by item type (Risk / Opportunity).
FR-RISK-06	The system shall allow a Risk/Opportunity to be tagged with the owning OBS node and, via the RACSI matrix, with a named accountability structure for its mitigation plan.

3.15 RACSI Accountability Matrix

A Responsible/Accountable/Consulted/Support/Informed matrix assigning OBS-sourced roles or named people to each of the seven NCP process steps and to selected governance items, with full CRUD via RBAC.

ID	Requirement
FR-RACSI-01	The system shall support full CRUD on RACSI Activities, each with a code, title, description, a reference type (NCP Process Step / Business Rule / Control / Risk-Opportunity / General), and an owning OBS node.
FR-RACSI-02	The system shall allow a RACSI Activity referencing the NCP process to be tagged with exactly one stage (S1-S7).
FR-RACSI-03	The system shall allow a RACSI Activity to be linked directly to a specific Business Rule, Control, or Risk/Opportunity record, so governance accountability is traceable to a concrete item rather than stated abstractly.
FR-RACSI-04	The system shall support full CRUD on RACSI Assignments against an Activity, each assigning exactly one of the five RACSI types (Responsible, Accountable, Consulted, Support, Informed) to exactly one assignee.
FR-RACSI-05	The system shall allow a RACSI Assignment's assignee to be given either as a Role (an org-wide accountability drawn from the RBAC role catalog) or as a specific Named Person (a user), sourced from the Organizational Breakdown Structure.
FR-RACSI-06	The system shall enforce that a RACSI Activity has at most one Accountable (A) assignment at any time, both in the create-assignment form (the "A" option is disabled once an Accountable exists) and at the data-persistence layer (a database constraint); a request that would create a second Accountable shall be rejected with HTTP 409.
FR-RACSI-07	The system shall allow a RACSI Activity's Responsible, Consulted, Support, and Informed types to each carry zero, one, or multiple assignments, freely mixing Role-based and Named-Person-based assignees.
FR-RACSI-08	The system shall allow an authorized user to remove an individual RACSI Assignment without deleting the parent Activity.
FR-RACSI-09	The system shall allow the RACSI Activity list to be filtered by reference type (NCP Process Step / Business Rule / Control / Risk-Opportunity / General).

3.16 Help & User Guide

An in-app, searchable, multi-language reference covering every module, available to every authenticated user regardless of role.

ID	Requirement
FR-HELP-01	The system shall provide an in-app Help screen presenting a set of topic articles covering: getting started, the S1-S7 NCP process, Hierarchy & OBS, Identity & RBAC, Governance/Risk/Compliance, the RACSI matrix, the AI Use Case Library & Governance, the Capitalization Library & AI agents, BPMN process diagrams, the AI Assistant, Alerts & Reports, and a FAQ.
FR-HELP-02	The system shall render Help content in the user's currently selected display language (English, French, or Arabic), matching the article set 1:1 across languages.
FR-HELP-03	The system shall provide a free-text search box on the Help screen that filters topics by a case-insensitive match against the topic title or body.

ID	Requirement
FR-HELP-04	The system shall make the Help screen accessible to every authenticated user independent of role or permission grants, consistent with the Dashboard.
FR-HELP-05	The system shall present each Help topic as an expandable/collapsible section so a user can scan titles before reading full content.

3.17 Alerts & Notifications

The ten standard alerts (A-J) from the Knowledge Base, computed by the Monitoring & Alert Agent.

ID	Requirement
FR-ALT-01	The system shall implement all ten standard alerts (A: new high-priority problem, B: immediate action overdue, C: corrective action pending evaluation, D: RCA not started, E: capitalization stage reminder, F: KPI threshold breach, G: immediate action due tomorrow, H: corrective action due in 2 days, I: evaluation due in 2 days, J: priority-1 sheet inactive 3+ days).
FR-ALT-02	The system shall provide an on-demand "Run Monitoring Agent" action, restricted to users holding alert.manage, that (re-)evaluates all alert conditions.
FR-ALT-03	The system shall avoid duplicate alerts of the same type against the same triggering entity within the same day.
FR-ALT-04	The system shall allow a user to mark an individual alert as read.
FR-ALT-05	The system shall display a live unread-alert badge in the navigation sidebar, refreshed periodically.

3.18 Reports & KPIs

The four standard reports and ten core KPIs from the Knowledge Base.

ID	Requirement
FR-REP-01	The system shall provide an Operational NCP Dashboard report listing every open/in-progress Sheet with stage, ageing in days, and immediate/corrective action completion percentages.
FR-REP-02	The system shall provide an Action Plan Monitoring report listing every action with owner, planned date, status, and an overdue flag.
FR-REP-03	The system shall provide a Strategic Problem-Solving Scorecard showing KPIs 2, 5, 7, 8, and 10, a priority distribution chart, and a non-conformity-by-department chart.
FR-REP-04	The system shall provide a Capitalization & Lessons Learned Log report listing every closed Sheet's lessons learned, standardization, and generalization status.
FR-REP-05	The system shall compute all 10 KPIs (see Appendix B) in real time from current data, scoped to the requesting user's tenant.

3.19 Multi-Language & Localization

English, French, and Arabic, with full right-to-left (RTL) layout for Arabic.

ID	Requirement
FR-I18N-01	The system shall provide complete UI translation in English, French, and Arabic for all navigation, labels, form fields, statuses, and messages.
FR-I18N-02	The system shall allow a user to change their display language at any time from the top navigation bar, without reloading lost work.
FR-I18N-03	The system shall persist a user's language preference and apply it automatically on subsequent logins.
FR-I18N-04	The system shall render the entire application right-to-left, including navigation, forms, and tables, when Arabic is selected.
FR-I18N-05	The system shall allow an Organization to configure a tenant-level default language.

3.20 Multi-Tenancy

Complete data isolation between Organizations sharing the same application instance.

ID	Requirement
FR-TEN-01	The system shall scope every data query by the authenticated user's organization_id; no API response shall ever include another Organization's data.
FR-TEN-02	The system shall isolate RBAC configuration, Capitalization Library content, and RAG retrieval per Organization.
FR-TEN-03	The system shall allow multiple independent Organizations, and Organizations grouped under a single Group entity, to coexist on one deployment.

3.21 Audit Trail

An immutable record of every state-changing action, for compliance and traceability.

ID	Requirement
FR-AUD-01	The system shall write an audit log entry for every CREATE, UPDATE, and DELETE operation on a major entity, capturing actor, timestamp, entity type/id, and before/after values.
FR-AUD-02	The system shall make the audit trail viewable, read-only, to users holding audit.view (e.g., the Auditor role).
FR-AUD-03	The system shall ensure audit log writes never block or fail the primary operation they accompany.

3.22 BPMN Process Modeling

Full CRUD, RBAC-gated management of standard BPMN 2.0 process diagrams, rendered and edited in-browser with a genuine BPMN modeler (bpmn.io's bpmn-js), seeded with a diagram of the NCP Solver process itself.

ID	Requirement
FR-BPMN-01	The system shall support full CRUD on BPMN Diagrams, each with a code, title, description, a BPMN 2.0 XML payload, and an optional owning OBS node.

ID	Requirement
FR-BPMN-02	The system shall render a stored BPMN Diagram using a standards-compliant BPMN 2.0 modeler/viewer component, not a static image.
FR-BPMN-03	The system shall grant read-only pan/zoom viewing of a BPMN Diagram to any user holding bpmn.view.
FR-BPMN-04	The system shall grant the full editing palette (tasks, gateways, events, sequence flows) and a Save Diagram action only to users holding bpmn.edit.
FR-BPMN-05	The system shall allow a new BPMN Diagram to be created starting from a blank canvas with a single Start Event, for users holding bpmn.create.
FR-BPMN-06	The system shall gate deletion of a BPMN Diagram behind bpmn.delete.
FR-BPMN-07	The system shall be seeded, per Organization, with a BPMN Diagram modeling the NCP Solver process itself end-to-end (S1 through S7), including the effectiveness-evaluation loop from S6 back to S5 when a corrective action is verified Not Effective.

3.23 AI Assistant (Data Query & Application Help)

A chat-style Q&A screen, open to every authenticated user (assistant.view is granted to all 9 standard roles), offering two independent, rule-based agents: a permission-scoped data query agent, and an application-help query agent grounded in a static FAQ knowledge base. Both modes are cataloged in the AI Use Case Library (section 3.11) at Production/Assistive, but neither requires an AI Use Case to be activated to run, and neither places a live call to an external LLM — the safer deterministic/RAG path is used by design, since a live model risks asserting a number or fact not present in the tenant's own data.

ID	Requirement
FR-ASST-01	The system shall provide a "Query My Data" mode that answers natural-language questions about the requesting user's own organization by matching the question, via keyword scoring, to one of a fixed set of data-query intents (as of this version: open Sheet count, closed Sheet count, overdue actions, the requesting user's own open actions, unread alert count, KPI/closure-rate summary, open risks ranked by score, controls not confirmed effective, active Business Rule count, active AI Use Case count, and "who is Accountable for <process step>").
FR-ASST-02	The system shall associate every data-query intent with the single RBAC permission required to answer it (e.g., riskOpportunity.view for risk questions, control.view for control questions), and shall evaluate that permission after matching the intent but strictly before running any query.
FR-ASST-03	When the requesting user's role lacks the permission required by the matched intent, the system shall return an explicit, human-readable refusal naming the missing permission, and shall not execute the underlying query or expose any of the underlying data.
FR-ASST-04	The system shall scope every data-query intent to the requesting user's own organization_id, consistent with FR-TEN-01; no data-query answer shall include another tenant's data.
FR-ASST-05	When no data-query intent scores above zero for a submitted question, the system shall return a message listing the categories of question it can answer, rather than an empty or misleading result.
FR-ASST-06	The system shall provide an "Ask About the Application" mode that answers "Can it...?" / "How do I...?" questions by ranking a static FAQ corpus with the same TF-IDF retrieval engine (RAG) used by the Capitalization Library, and returning the top matching question/answer pairs with a similarity score.

ID	Requirement
FR-ASST-07	The system shall maintain the application-help FAQ corpus as a versioned, in-code data file covering, at minimum, every module documented in section 3 of this specification.
FR-ASST-08	The system shall log every AI Assistant invocation (mode, question, matched intent or top result score, answer, confidence) to the AI Agent Log, consistent with FR-AI-01.
FR-ASST-09	The system shall gate both AI Assistant modes behind a single assistant.view permission, independent of and in addition to any module-specific permission a given data-query intent may separately require.

4. External Interface Requirements

4.1 User Interfaces

The application presents the following screens, each accessible only when the authenticated user holds the corresponding view permission; navigation itself is filtered by permission (role-based menu visibility).

Screen	Purpose
Login	Authenticate with email/password; language selector; demo-account picker.
Dashboard	Role-aware summary: KPI tiles, recent Sheets, unread alerts, department chart.
NCP Sheets (list)	Filterable/searchable list of all Sheets in scope.
New NCP Sheet	S1 registration form.
NCP Sheet Detail	7 tabs, one per stage S1-S7, each with stage-appropriate AI assistance.
My Actions	Actions owned by, or pending evaluation by, the current user.
Capitalization Library	RAG semantic search over closed Sheets.
Standards	CRUD on the standards knowledge base.
AI Use Case Library (list)	Card grid of 14 AI use cases with tier/maturity/status badges, an Organization Active/Inactive toggle per card, and the browser-local Real LLM Provider Connection panel.
AI Use Case Detail	Metadata editor (tier, module, trigger, output, human checkpoint), current-version labeled-section editor, version history with revert, Organization activation card, per-Project override list, and the AI Usage Log.
Business Rules	CRUD on process business rules.
Controls (COSO)	CRUD on controls, filterable by COSO component.
Risks & Opportunities	Likelihood x impact heat-map matrix + CRUD register.
RACSI Matrix	Per-activity R/A/C/S/I assignment manager (roles or named people), filterable by reference type.
BPMN Process (list)	CRUD list of BPMN diagrams, each opening into a viewer or full modeler depending on RBAC.
BPMN Diagram	In-browser BPMN 2.0 viewer (pan/zoom, read-only) or modeler (full editing palette + Save), selected by permission.
AI Assistant	Two-mode chat: Query My Data (RBAC-scoped) and Ask About the Application (FAQ RAG search), open to every role.
Reports	4 standard reports in a tabbed view.

Screen	Purpose
Alerts	Alert inbox with manual "Run Monitoring Agent" trigger.
Hierarchy	Group / Organization / Project management.
Organizational Breakdown (OBS)	Site/department/service/team tree management, with live linked-item counts per node.
Users & Scope	User directory and role assignment, scoped to an OBS node.
Permission Matrix	Full role x permission grid editor.
Governance Settings	KPI thresholds, alert toggles, RCA default, REX policy.
License & Plan	Plan tier, deployment model, seats, billing.
Help	Searchable, multi-language user guide covering every module; open to all authenticated users.

4.2 Software Interfaces (REST API)

All interfaces are exposed as a JSON REST API under /api, secured by a JWT bearer token (except /api/auth/login), and enforced per-endpoint by the RBAC permission catalog. The API is grouped by module as follows.

Module	Endpoints
Auth	POST /api/auth/login, GET /api/auth/me, PUT /api/auth/me/language
Hierarchy	GET /api/hierarchy/tree, PUT /api/hierarchy/organization, POST /api/hierarchy/organizations, GET|POST|PUT|DELETE /api/hierarchy/projects
OBS	GET|POST /api/obs, PUT|DELETE /api/obs/:id
Users	GET|POST /api/users, GET|PUT|DELETE /api/users/:id, PUT /api/users/:id/roles, GET /api/users/directory
Roles & Permissions	GET|POST /api/roles, PUT|DELETE /api/roles/:id, GET /api/roles/directory, GET /api/roles/permissions/catalog, GET /api/roles/permissions/matrix, PUT /api/roles/:id/permissions
License	GET|PUT /api/license
Governance	GET|PUT /api/governance
Standards	GET|POST /api/standards, GET|PUT|DELETE /api/standards/:id
NCP Sheets	GET|POST /api/fiches, GET|PUT|DELETE /api/fiches/:id, POST /api/fiches/:id/transition, POST /api/fiches/:id/close, PUT /api/fiches/:id/understanding, PUT /api/fiches/:id/team, GET|POST|PUT|DELETE /api/fiches/:id/root-causes[/:rcId], PUT /api/fiches/:id/rex, POST /api/fiches/:id/rex/generate-draft
Actions	GET|POST /api/actions, PUT /api/actions/:id, PUT /api/actions/:id/progress, PUT /api/actions/:id/evaluation, GET|POST /api/actions/:id/evidence

Module	Endpoints
AI Agents (process)	POST /api/ai-agents/:ficheId/classification, /problem-structuring, /containment-advisor, /root-cause-mining, /action-recommendation, /evaluation-assistant, GET /api/ai-agents/:ficheId/logs
Capitalization	GET /api/capitalization, GET /api/capitalization/search
Alerts	GET /api/alerts, POST /api/alerts/run, PUT /api/alerts/:id/read
AI Use Cases	GET|POST /api/ai-use-cases, GET|PUT|DELETE /api/ai-use-cases/:id, GET /api/ai-use-cases/providers, PUT /api/ai-use-cases/:id/activation, PUT /api/ai-use-cases/:id/project-override/:projectId, GET|POST /api/ai-use-cases/:id/versions, GET /api/ai-use-cases/:id/versions/:versionId, POST /api/ai-use-cases/:id/versions/:versionId/revert
AI Usage Log	GET /api/ai-usage-log (optional ?useCaseId=), POST /api/ai-usage-log — append-only, no PUT/PATCH/DELETE route exists
Business Rules	GET|POST /api/business-rules, GET|PUT|DELETE /api/business-rules/:id
Controls	GET|POST /api/controls, GET|PUT|DELETE /api/controls/:id
Risks & Opportunities	GET|POST /api/risks, GET|PUT|DELETE /api/risks/:id
RACSI	GET|POST /api/racsi, GET|PUT|DELETE /api/racsi/:id, POST /api/racsi/:id/assignments, DELETE /api/racsi/:id/assignments/:assignmentId
BPMN	GET|POST /api/bpmn, GET|PUT|DELETE /api/bpmn/:id
AI Assistant	POST /api/assistant/query-data, POST /api/assistant/query-app
Reports	GET /api/reports/kpis, /operational, /action-plan, /scorecard, /capitalization
Dashboard	GET /api/dashboard

4.3 Hardware Interfaces

None. NCP Solver is a standard web application with no direct hardware interface requirement; it runs on commodity server and client hardware.

4.4 Communications Interfaces

- HTTP/HTTPS using JSON request/response bodies for all client-server communication.
- Authentication via a signed JWT Bearer token in the Authorization header, issued at login and valid for 12 hours.
- The web client is served separately from the API and communicates with it exclusively over the REST interface (no server-side page rendering).

4.5 Interfaces with External Applications

NCP Solver is architected — modular services behind a stable REST/JSON API, tenant isolation, and the composable-module principle of NFR-MAINT-01 — to integrate with the external systems below. Each entry states, honestly, whether it is implemented in this version or architecture-ready for a future release; none of the architecture-ready interfaces performs a live external call today.

Identity Provider / Single Sign-On (SSO) — Architecture-ready

The current implementation authenticates locally (email + bcrypt-hashed password, issuing its own JWT). The RBAC layer already separates authentication (who is this user) from authorization (what can this role do), so a future release can add SAML 2.0 or OpenID Connect (OIDC) authentication against a corporate Identity Provider (e.g., Azure AD/Entra ID, Okta, Google Workspace) by replacing only the login/token-issuance step; every downstream permission check, and the JWT-bearer contract used by every other endpoint, is unaffected.

ERP / MES (Manufacturing & Enterprise Resource Planning) — Architecture-ready

No live ERP/MES connector is implemented in this version. The NCP Sheet, Action, and Standard entities are modeled with stable REST endpoints and tenant-scoped identifiers, so a future integration can push non-conformities detected in an ERP/MES quality module into NCP Solver (via POST /api/fiches) or pull closure status back out, without a schema change.

Document Management System (DMS) — Architecture-ready

Action evidence (execution/evaluation proof) and REX attachments are currently recorded as metadata records (ActionEvidence) rather than files stored in an external DMS (e.g., SharePoint, a corporate document vault). The ActionEvidence entity is deliberately a thin, replaceable pointer so a future version can store the underlying file in an external DMS and keep only its reference here.

SMTP / Email Notification Gateway — Architecture-ready

Alerts (section 3.17) are currently delivered only in-app (the sidebar unread-alert badge and the Alerts inbox). No outbound SMTP integration is implemented. The Alert & Notification Engine is one of the explicitly composable modules (NFR-MAINT-01); adding an email-delivery channel means adding a delivery adapter behind the existing NotificationAlert entity, not redesigning the alert logic itself.

Business Intelligence (BI) / Data Warehouse Export — Architecture-ready

The Reports & KPIs module (section 3.18) computes all 10 KPIs and 4 standard reports live from the relational schema. No scheduled export or BI-tool connector (e.g., Power BI, Tableau) is implemented in this version; the relational schema (section 6) is a stable, documented basis for a future read replica or scheduled extract.

Webhooks / Event Notifications to Other Systems — Architecture-ready

No outbound webhook mechanism is implemented in this version — other systems cannot currently subscribe to NCP Solver events (Sheet closed, Alert raised, Control tested). The Audit Trail (section 3.21) already captures every state-changing event with actor, timestamp, and before/after values, giving a ready event source a future webhook dispatcher could be built on.

LLM / AI Provider APIs — Implemented (optional, user-configured)

The AI Use Case Library's generation service (section 3.11) is the single, designated server-side integration point for a live LLM call, per NFR-MAINT-04. A user configures their own Real LLM Provider Connection entirely client-side (stored only in browser local storage, per FR-AIUC-17/18) and it is forwarded once per generation request; the server places exactly one outbound call for that request and never persists the connection (NFR-SEC-08). Five providers have a working native REST implementation in this version — Anthropic, OpenAI, Azure OpenAI, Mistral, and a self-hosted/local Ollama endpoint, plus any OpenAI-compatible Custom endpoint — while Google, AWS Bedrock, Cohere, and Meta Llama are listed in the provider catalog but not yet wired to a live call. Every AI Use Case always computes its deterministic, RAG-grounded result first (per NFR-DATA-01) and falls back to it automatically on any live-call failure, so this integration is a supplementary enhancement, never a dependency.

Sibling Enterprise Applications (Risk Management, Audit Management, Project Management) — Architecture-ready

Per section 2.1 and NFR-MAINT-01, the Action Register, Alert & Notification Engine, User Management & RBAC layer, KPI/Reporting engine, and Audit Trail Service are built as composable modules behind the same REST/JSON API contract used internally, so a sibling application can call them directly rather than reimplementing equivalent functionality. No such external application is connected in this version; the interface is the existing, documented API (section 4.2).

5. Non-Functional Requirements

5.1 Performance

ID	Requirement
NFR-PERF-01	Dashboard and list views shall render within 2 seconds under nominal load (per-tenant data volumes as seeded: dozens of Sheets, hundreds of actions).
NFR-PERF-02	AI agent suggestions (classification, containment, root-cause mining, action recommendation) shall return within 5 seconds.
NFR-PERF-03	Capitalization Library search and AI Assistant application-help search shall each return ranked results within 1 second for a tenant's full corpus.

5.2 Security

ID	Requirement
NFR-SEC-01	All API endpoints except login shall require a valid JWT bearer token.
NFR-SEC-02	Passwords shall be stored only as bcrypt hashes; plaintext passwords shall never be logged or persisted.
NFR-SEC-03	Every authorization decision shall be enforced server-side; client-side permission checks are a UX convenience only, never the security boundary.
NFR-SEC-04	No SQL query shall be constructed by string-concatenating untrusted input; all data access shall use parameterized statements.
NFR-SEC-05	AR/AE segregation of duties (FR-ID-07) shall be enforced server-side, not merely hidden in the UI.
NFR-SEC-06	The one-Accountable-per-RACSI-activity rule (FR-RACSI-06) shall be enforced by a database-level constraint, not merely by disabling the option in the UI.
NFR-SEC-07	A Real LLM Provider Connection's API key (FR-AIUC-17) shall be captured and stored only in the user's browser local storage; the backend shall never persist it to any table, write it to any log line, or return it in any API response.
NFR-SEC-08	When a generation request forwards a Real LLM Provider Connection, the server shall use it for exactly one outbound call to the configured provider and shall never write, log, cache, or hold it beyond that single request; an optional server-side fallback API key, if configured via environment variable, shall apply only to the AI Assistant's data-query and application-help modes (section 3.23), never to the AI Use Case Library's own generation calls, which always prefer the user's own browser-stored connection.

5.3 Availability & Reliability

ID	Requirement
NFR-AVAIL-01	The system shall degrade gracefully if an AI agent call fails: the underlying CRUD operation shall still succeed, with the AI suggestion simply unavailable.

ID	Requirement
NFR-AVAIL-02	The system shall target 99.9% uptime in a production deployment.

5.4 Scalability

ID	Requirement
NFR-SCALE-01	The architecture shall support horizontal scaling of the web tier independently of the API tier.
NFR-SCALE-02	The system shall support at least 100 concurrent users per tenant without functional degradation.
NFR-SCALE-03	The relational schema and RAG index shall each be swappable for a distributed equivalent (e.g., managed PostgreSQL, a vector database) without changing the API contract.

5.5 Usability & Accessibility

ID	Requirement
NFR-UX-01	The UI shall follow the POWERACT Consulting visual identity (color palette, typography, iconography) consistently across all modules.
NFR-UX-02	The UI shall be responsive and usable on desktop and tablet viewport widths.
NFR-UX-03	All interactive controls shall be keyboard-navigable and carry accessible labels (WCAG 2.1 AA target).

5.6 Maintainability & Portability

ID	Requirement
NFR-MAINT-01	The Action Register, Alert & Notification Engine, User Management & RBAC layer, KPI/Reporting engine, and Audit Trail Service shall be implemented as composable modules reusable by other applications (Risk Management, Audit Management, Project Management) via the same API contract.
NFR-MAINT-02	The backend shall run on any standard Node.js 22.5+ runtime without requiring a dedicated database server, container orchestration, external API keys, or a native/compiled build toolchain (the database layer uses Node's built-in node:sqlite module).
NFR-MAINT-03	The AI agent implementations shall be isolated behind a stable internal interface so a rule-based implementation can be replaced by an LLM/vector-DB-backed one without changing calling code.
NFR-MAINT-04	The AI Use Case Library's generation service (section 3.11) shall be the single designated server-side integration point for a live LLM call and its provider-specific credential and endpoint logic; no other module shall embed its own LLM provider client.

5.7 Auditability & Compliance

ID	Requirement
NFR-COMP-01	Every AI-generated suggestion shall be logged with a confidence score and be attributable to a specific agent and Sheet (or, for the AI Assistant, to the requesting user and question), per the AIAgentLog entity.
NFR-COMP-02	The Controls module shall support demonstrating COSO Internal Control - Integrated Framework coverage across all five components.
NFR-COMP-03	The system shall retain the full version history of every AI Use Case indefinitely (no hard deletion of prior versions on edit or revert).

5.8 Data Retention & Privacy

ID	Requirement
NFR-DATA-01	By default, and whenever no Real LLM Provider Connection is configured, the system shall not transmit tenant data to any third-party service; every AI Use Case shall run entirely within the deployment's own infrastructure using rule-based/statistical methods and RAG grounding. Only when a user has explicitly configured and forwarded their own browser-stored Real LLM Provider Connection (FR-AIUC-17) may that single generation request be sent to that user-selected third-party provider; this is opt-in per browser, never enabled by default, and never applies to the AI Assistant's data-query mode.
NFR-DATA-02	The system shall support both SaaS (shared infrastructure, tenant-isolated data) and OnPrem (customer-hosted) deployment models per Organization.

6. Data Model Overview

The relational schema comprises the following core entities (full attribute-level detail is maintained in the Information Model specification and the implementation's schema.sql). All entities carry an organization_id (tenant) foreign key except Group and the global Permission catalog.

Entity	Description
Group	Optional holding entity above Organization; groups sibling Organizations (e.g., a group of companies).
Organization	The tenant boundary. Holds sector, country, and all tenant-isolated data.
Project	Optional sub-entity of an Organization; a Sheet may reference one.
ObsNode	A node in the Organizational Breakdown Structure tree (site/department/service/team); referenced by users, Sheets, Business Rules, Controls, Risks/Opportunities, RACSI Activities, and BPMN Diagrams as their owning org unit.
User	A person; holds RBAC roles; tenant-scoped.
Role	A named set of permissions, tenant-scoped; 9 system templates + custom roles.
Permission	A global catalog entry (module.action) granted to roles.
License	Plan tier, deployment model, seats, billing per Organization.
GovernanceSettings	KPI thresholds, alert toggles, RCA default, REX policy per Organization.
Standard	A normative document (ISO standard, internal procedure).
NCPFiche (NCP Sheet)	The central aggregate: a non-conformity/problem record spanning S1-S7.
NCPTeamAssignment	Join entity: users assigned to a Sheet's team.
ProblemUnderstanding	5W2H structured analysis, one per Sheet.
RootCause	A 6M-categorized cause identified in S4.
Action	Abstract base for Immediate and Corrective actions; Action Owner (AR), dates, status.
ActionEvaluation	Effectiveness verdict for an Action; enforces AR != AE.
ActionEvidence	File attachment evidencing execution or evaluation of an Action.
REXEntry	S7 capitalization content: lessons learned, standardization/generalization.
NotificationAlert	An instance of one of the 10 alert types (A-J).
AIAgentLog	Immutable log of every AI agent invocation with confidence score, including AI Assistant queries.
AuditLog	Immutable log of every CREATE/UPDATE/DELETE on a major entity.
AIUseCase	Metadata for an AI use case (stable across versions): tier (Assistive/Augmented), owning module key, trigger/output/human-checkpoint descriptions, and its Organization-level

Entity	Description
	Active/Inactive flag.
AIUseCaseVersion	A versioned, labeled content snapshot of an AI use case; append-only.
AIUseCaseProjectOverride	A per-Project, tri-state (Inherit/On/Off) activation override on an AI Use Case; deleted automatically (cascade) when its AI Use Case is deleted.
AIUsageLog	Append-only record of one AI suggestion outcome (Accepted/Edited/Rejected), linking the AI Use Case, Organization, Project, acting user, and an output summary; deleted automatically (cascade) when its AI Use Case is deleted.
BusinessRule	A governed IF/THEN process rule with severity, owner, and owning OBS node.
Control	A COSO-classified internal control with type, frequency, effectiveness, and owning OBS node.
RiskOpportunity	A risk or opportunity item with likelihood/impact scoring and owning OBS node.
RiskControl	Join entity: controls mitigating a given risk/opportunity.
RacsiActivity	One of the 7 NCP process steps or a governance item linked to a Business Rule/Control/Risk record; carries an owning OBS node.
RacsiAssignment	One R/A/C/S/I assignment on a RacsiActivity, pointing to either a Role or a User; at most one Accountable per Activity.
BpmnDiagram	A stored BPMN 2.0 process diagram (code, title, description, XML payload), owned by an OBS node; rendered/edited in-browser.

Appendix A — Permission Catalog Summary

The full permission matrix (role x permission) is administered live in the application's Permission Matrix screen and comprises 80 permission codes across 24 modules (dashboard, help, fiche, action, rootcause, rex, standard, capitalization, user, role, hierarchy, obs, license, governance, aiUseCase, businessRule, control, riskOpportunity, racsi, bpmn, assistant, report, alert, audit), each with up to five actions (view, create, edit, delete, and module-specific actions such as validate, close, assignTeam, updateOwn, evaluate, manageRoles, managePermissions, activate, projectOverride, viewUsageLog, manage, export). Section 2.3 summarizes the resulting access per standard role; the live matrix is the system of record.

Appendix B — KPI Formulas

KPI	Name	Formula	Target
KPI1	Completion Rate - Immediate Actions	(AI Done / AI Total) x 100	100%
KPI2	Effectiveness Rate - Immediate Actions	(AI Effective / AI Done) x 100	>= 85%
KPI3	On-Time Completion Rate	(Actions on time / Total done) x 100	>= 80%
KPI4	Completion Rate - Corrective Actions	(AC Done / AC Total) x 100	100%
KPI5	Effectiveness Rate - Corrective Actions	(AC Effective / AC Done) x 100	>= 80%
KPI6	Completion Rate - Evaluations	(Evals Done / Evals Due) x 100	100%
KPI7	Standardization Rate	(Sheets w/ std. update / Closed) x 100	>= 70%
KPI8	Generalization Rate	(Sheets generalized / Closed) x 100	>= 40%
KPI9	Non-Conformity Count per Department	Count of Sheets per department per period	Trending down
KPI10	Sheet Closure Rate	(Closed / Total Opened) x 100	>= 85%

Appendix C — Alert Catalog (A-J)

Alert	Name	Trigger	Target
A	New High-Priority Problem	New Sheet with criticality High or priority 1	CI Pilot + NCP Team
B	Immediate Action Overdue	Planned date passed, status not Done	AR / CI Pilot
C	Corrective Action Pending Evaluation	Marked Done, or evaluation due in 2 days	Evaluator (AE)
D	RCA Not Started	48h after detection, no root cause recorded	CI Pilot
E	Capitalization Stage Reminder	All corrective actions Effective	NCP Team
F	KPI Threshold Breach	Monthly KPI below configured threshold	Direction + CI Pilot
G	Immediate Action Due Tomorrow	Planned date = tomorrow	Action Owner (AR)
H	Corrective Action Due in 2 Days	Planned date in 2 days	Action Owner (AR)
I	Evaluation Due in 2 Days	Planned review date in 2 days	Evaluator (AE)
J	Priority-1 Sheet Inactive 3 Days	No update for 3+ days on a P1 Sheet	CI Pilot

Appendix D — Glossary

Term	Definition
NCP Sheet	The central record of a non-conformity or problem, spanning stages S1-S7 (referred to as "NCP Fiche" in the original source specification; stages were renumbered from E1-E7 to S1-S7 in the delivered implementation).
S1-S7	The seven stages of the NCP resolution process: Detection & Alert, Problem Understanding, Immediate/Containment Actions, Root Cause Analysis, Corrective Action Plan, Execution & Evaluation, Capitalization (REX).
AR	Action Responsible (Action Owner) - the user responsible for executing an action. Renamed from RR (Responsable Réalisation) in the delivered implementation.
AE	Action Evaluator - the user responsible for verifying an action's effectiveness; must differ from the AR. Renamed from RE (Responsable Évaluation) in the delivered implementation.
REX	Retour d'EXperience - the structured lessons-learned output of stage S7.
RCA	Root Cause Analysis - the discipline of finding the true underlying cause of a problem (5-Why, Ishikawa).
5W2H / QQOQCCP	Structured problem-description framework: What/Where/When/Who/Why/How/How-Much (French: Qui/Quoi/Ou/Quand/Comment/Combien/Pourquoi).
6M	Ishikawa cause categories: Man, Machine, Method, Material, Measurement, Milieu.
RAG	Retrieval-Augmented Generation - TF-IDF/cosine-similarity semantic search grounding AI suggestions and the AI Assistant's application-help answers in the tenant's own historical data or the built-in FAQ corpus, without a live external LLM call.
RBAC	Role-Based Access Control - access rights determined by assigned role(s).
COSO	Committee of Sponsoring Organizations of the Treadway Commission - the internal control framework used to classify Controls.
OBS	Organizational Breakdown Structure - the site/department/service/team tree; the shared source of roles and named people that other modules, including RACSI, point back to.
RACSI	Responsible, Accountable, Consulted, Support, Informed - a five-role extension of the classic RACI accountability model; exactly one Accountable per activity, the other four roles allow multiple assignees.
BPMN	Business Process Model and Notation - the OMG standard notation used by the BPMN Process module to model workflows, including the NCP Solver process itself.
LLM	Large Language Model - the class of AI model an AI Use Case may optionally call via a user's own Real LLM Provider Connection (section 3.11), stored only in that user's browser and forwarded once per generation request; every AI Use Case still computes its deterministic result and falls back to it automatically if the live call fails.
AI Assistant	The chat-style Q&A feature (section 3.23) offering RBAC-scoped data queries and application-help queries, distinct from both the per-stage AI Agents and the AI Use Case Library.

Term	Definition
Assistive / Augmented	The two AI Use Case tiers (section 3.11): Assistive means the AI only observes/analyzes/suggests while a human performs the task and decision; Augmented means the AI performs a substantial part of the task but a human must review, edit, and approve before it is finalized. A third tier, Autonomous, is explicitly out of scope.
Multi-Tenant	Architecture in which multiple Organizations share one deployment with fully isolated data.
Composable Module	A module (Action Register, Alert Engine, RBAC, Audit Trail) built for reuse by other applications via a shared API contract.